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Factors associated with early cyclicity in postpartum dairy cows

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ABSTRACT

The objective of this study was to evaluate factors associated with resumption of ovarian cyclicity within 21 days in milk (DIM) in dairy cows. Cows ($n = 768$) from 2 herds in north Florida had their ovaries scanned at 17 ± 3 , 21 ± 3 , and 24 ± 3 DIM. Cows that had a corpus luteum ≥ 20 mm at 17 ± 3 or at 21 ± 3 DIM or that had a corpus luteum < 20 mm in 2 consecutive examinations were determined to be cyclic by 21 ± 3 DIM. The following information was collected for up to 14 DIM: calving season, parity, calving problems, metabolic problems, metritis, mastitis, digestive problems, lameness, body weight loss, dry period length, and average daily milk yield. Body condition was scored at 17 ± 3 DIM. Multivariable mixed logistic regression analysis was performed using the GLIMMIX procedure of SAS. Variables with $P \leq 0.2$ were considered in each model. Herd was included as a random variable. Three models were constructed: model 1 included all cows, model 2 included only cows from dairy 1 that had daily body weights available, and model 3 included only multiparous cows with a previous dry period length recorded. In model 1, variables associated with greater cyclicity by 21 ± 3 DIM were calving in the summer and fall rather than in the winter or spring, being multiparous rather than primiparous, and not having metabolic or digestive problems. In model 2, variables associated with greater cyclicity by 21 ± 3 DIM were calving in the summer and fall, not having metritis or digestive problems and not losing > 28 kg of BW within 14 DIM. In model 3, variables associated with greater cyclicity by 21 ± 3 DIM were absence of metabolic problems and dry period ≤ 76 d. In summary, cyclicity by 21 ± 3 DIM was negatively associated with calving in winter

or spring, primiparity, metritis, metabolic or digestive problems, loss of > 28 kg of body weight, and a dry period > 76 d. Strategies preventing extended dry period length and loss of BW, together with reductions in the incidence of metritis as well as metabolic and digestive problems should improve early cyclicity postpartum.

Key words: risk factor, early cyclicity, dairy cow

INTRODUCTION

Reproductive performance is one of the main determinants of profitability in dairy farms (De Vries, 2006; Galvão et al., 2013). Resumption of ovarian cyclicity before the first AI (Galvão et al., 2004; Chebel et al., 2006; Santos et al., 2009), but especially cyclicity earlier in lactation (Darwash et al., 1997; McCoy, 2006), is associated with improved reproductive performance. Two recent studies in North America found that Holstein cows that were cyclic by 21 DIM had improved reproductive performance compared with cows that cycled later in lactation or cows that were not cyclic by 50 to 60 DIM (Galvão et al., 2010a; Dubuc et al., 2012).

Virtually all Holstein cows have the first wave of follicle growth starting 2 wk postpartum, with approximately 30 to 40% of these cows ovulating between 16 and 20 DIM; another 30 to 40% of the cows will ovulate follicles from subsequent follicular waves from 30 to 50 DIM and approximately 20 to 40% remain anovulatory by 50 to 60 DIM (Butler, 2003; Santos et al., 2009; Galvão et al., 2010a). Few studies have evaluated risk factors that affect cyclicity early postpartum. A recent study focused on the effect of indicators of energy balance (i.e., BHBA and NEFA) and indicators of inflammation (i.e., haptoglobin) on cyclicity at 21 ± 3 DIM (Dubuc et al., 2012). Those authors reported that cows with increased NEFA pre- and postpartum had decreased cyclicity at 21 ± 3 DIM. The association between negative energy balance and resumption of ovarian cyclicity has been well described (Butler, 2003). Deep negative energy balance affects LH pulsatility and

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blood glucose, insulin, and IGF-1 concentrations, which ultimately affect follicle development, estradiol production, and ovulation (Canfield and Butler, 1990; Beam and Butler, 1999; Butler, 2003).

Dubuc et al. (2012) also showed that cows with increased haptoglobin postpartum had decreased cyclicity at 21 ± 3 DIM. Metritis is the most common disease associated with increased haptoglobin early postpartum (Huzzey et al., 2009; Galvão et al., 2010b). In a previous study, it was observed that cows that were cyclic at 21 ± 3 DIM had decreased prevalence of cytological endometritis at 49 ± 3 DIM compared with cows that were not cyclic by 49 ± 3 DIM (Galvão et al. 2010a). Metritis is a major risk factor for endometritis (Benzaguen et al., 2007; Cheong et al., 2011). These findings indicate that metritis might be an important factor associated with early ovulation. Gram-negative bacterial LPS concentrations are increased in the uterine fluid (Mateus et al., 2003), plasma (Herath et al., 2009), and follicular fluid (Herath et al., 2007) when cows have a uterine infection. Large quantities of LPS are also released during coliform mastitis (Carroll et al., 1964; Hill et al., 1979) and SARA (Khafipour et al., 2009), which are common conditions in early postpartum. Lipopolysaccharide has been shown to impair the release of both GnRH and LH (Peter et al., 1989; Rivest et al., 1993; Battaglia et al., 1999) and to decrease aromatase activity (Herath et al., 2009), which ultimately may affect follicular growth, estradiol production (Williams et al., 2007, 2008), and decrease ovulation rate (Peter et al., 1989). Dubuc et al. (2012) also observed that calving season and parity were associated with early cyclicity postpartum; however, other postpartum conditions such as calving problems (e.g., dystocia, twins, stillbirths or retained placenta), infectious diseases such as mastitis or lameness, and metabolic diseases such as hypocalcemia and ketosis may also directly, through inflammation, or indirectly, through negative energy balance, affect cyclicity early postpartum. Other factors that may be important for early ovulation because of their association with negative energy balance are BW loss and dry period length (Opsomer et al., 2000; Shrestha et al., 2005). Therefore, the objective of our study was to identify risk factors associated with resumption of ovarian cyclicity early postpartum (cyclicity by 21 ± 3 DIM).

MATERIALS AND METHODS

Cows, Housing, and Feeding

Seven hundred sixty-eight cows (280 primiparous and 488 multiparous; 461 cows from dairy 1 and 307 cows from dairy 2) from 2 freestall dairies located in

north-central Florida were used for the current study. Cows calving from December 2010 through August 2012 were enrolled at 17 ± 3 DIM. Our study included control cows ($n = 245$) from a previous study (Bittar et al., 2014). All animal procedures were approved by the University of Florida Institutional Animal Care and Use Committee.

Dairy 1, the University of Florida Dairy Unit (Gainesville), milked approximately 500 Holstein cows twice daily with a rolling herd average of approximately 10,000 kg/cow. The freestall beds and walking alleys were cleaned twice daily. Twice weekly, clean and dry sand was added on the top of the freestall beds. Fans with misters and sprinklers over the feed line were present in the barns and activated when environmental temperatures rose above 18°C . Dairy 2, a privately owned dairy, milked approximately 4,800 Holstein cows thrice daily with a rolling herd average of approximately 10,500 kg/cow. The freestall beds and walking alleys were cleaned thrice daily. Twice weekly, clean and dry sand was added on the top of the freestall beds. Cows were housed in tunnel-ventilated freestall barns equipped with misters that were activated when environmental temperatures increased above 18°C . In both dairies, cows were fed twice daily a TMR to meet or exceed the requirements of a second lactation Holstein cow weighing 680 kg of BW and producing 45 kg of 3.5% FCM per day. Fresh water was available ad libitum. In both dairies primiparous and multiparous were housed separately. Cows were vaccinated and treated for common diseases according to standard operating procedures developed with participation of the veterinarians from University of Florida, College of Veterinary Medicine, Food Animal Reproduction, and Medicine Service.

Study Design, Sample Size, and Data Collection

This was a longitudinal cohort study. The data was conveniently collected from December 2010 to August 2012 or selected (control cows; $n = 245$) from a previous study as previously mentioned (Bittar et al., 2014). Because the data was sampled by convenience, no a priori calculation of sample size occurred. Therefore, to address this shortfall, power was calculated and presented in the statistical analysis. All cows had their ovaries examined by ultrasonography (US) at 17 ± 3 , 21 ± 3 , and 24 ± 3 DIM using a portable ultrasound scanner (Easi-Scan, 4.5- to 8.5-MHz linear transducer, BCF Technology Ltd., Livingston, UK). Cows that had a corpus luteum (CL) ≥ 20 mm at its longest diameter (McArt et al., 2010) at 17 ± 3 or at 21 ± 3 DIM were determined to be cyclic by 21 ± 3 DIM. Cows that had a CL < 20 mm at 17 ± 3 DIM were scanned again at

21 $\pm$ 3 DIM and cows that had a CL <20 mm at 21 $\pm$ 3 DIM were scanned again at 24 $\pm$ 3 DIM. Cows that had a CL <20 mm in 2 consecutive US examinations were also determined to be cyclic by 21 $\pm$ 3 DIM. Only 3 cows had a CL <20 mm observed at 17 $\pm$ 3 or 21 $\pm$ 3 DIM that were not confirmed at 21 $\pm$ 3 or 24 $\pm$ 3 DIM, respectively, and those cows were classified as not being cyclic by 21 $\pm$ 3 DIM. The outcome of interest was cyclicity by 21 $\pm$ 3 DIM considering that cows that were cyclic at 21 $\pm$ 3 DIM had increased fertility compared with cows that started cycling later in lactation (Galvão et al., 2010a).

All the cows had a complete physical examination at 4, 7, and 12 DIM in dairy 1 and at 4, 6, and 10 DIM in dairy 2 for diagnosis of common infectious (i.e., metritis and mastitis) and metabolic diseases [i.e., displaced abomasum (**DA**), and ketosis]. Body condition was determined based on a 5-point scoring system (Ferguson et al., 1994) at 17 $\pm$ 3 DIM. Furthermore, all cows were visually monitored daily and any cow showing signs of depression, inappetence, or lethargy was submitted to a complete physical examination. Both farms recorded daily milk yield and any cow with a decrease $\geq$ 10% in milk yield was automatically sorted for a physical examination using automatic sorting gates (DeLaval Sort Gate DSG3, DeLaval International AB, Tumba, Sweden). Any condition or disease diagnosed between calving and 14 DIM, such as abortion, stillbirth, dystocia, clinical hypocalcemia, ketosis, retained placenta (**RP**), metritis, mastitis, and lameness, were recorded on the farm management software and later retrieved and entered into an Excel spreadsheet (Microsoft Corp., Richmond, WA). Dairy 1 used the Afifarm software (Afikim Kibbutz, Afikim, Israel) and dairy 2 used PCDart (Dairy Records Management Systems, Raleigh, NC). Calving difficulty was scored using a scale from 1 to 5 (1 = no assistance; 2 = assistance by 1 person without the use of mechanical traction; 3 = assistance by 2 or more people; 4 = assistance with mechanical traction; 5 = fetotomy or Cesarean section). Stillbirth was defined as the birth of a dead calf or a calf that died within 24 h of birth. Retained placenta was characterized by failure to release the placenta within 24 h of parturition. Clinical hypocalcaemia was characterized by recumbency that was responsive to intravenous calcium administration. Ketosis was defined as presence of ketone bodies in urine which resulted in any color change [5 mg/dL (trace) or higher] in the urine test strip (Ketostix, Bayer, Pittsburgh, PA). The test strip has been reported to have 88% sensitivity (detected 29 out of 33 positive cows) and 95% specificity (detected 39 out of 41 negative cows) using blood BHBA concentrations $\geq$ 1.2 mmol/L as the gold standard (Galvão et al., 2012). Metritis was characterized by the presence

of red-brownish, watery fetid uterine discharge (Benzaquen et al., 2007). Mastitis was characterized by abnormal milk or mammary gland. Displaced abomasum was diagnosed by a characteristic ping over the 9th to 13th ribs and was confirmed during surgery. Indigestion was characterized by diarrhea, decreased rumen motility, or bloat. Lameness was characterized by favoring one of the limbs while walking and was confirmed during hoof trimming. Length of the dry period was collected for multiparous cows. Body weight was only available for cows in dairy 1.

Statistical Analyses

Cow was considered the experimental unit and the outcome variable was ovulation by 21 DIM. All predictors were dichotomized based on previous knowledge from the literature, biological relevance, or trends in the data set. Data were initially analyzed by Chi-square test using the FREQ procedure of SAS (Version 9.3, SAS Institute Inc., Cary, NC) and subsequently by multivariable mixed logistic regression models using the GLIMMIX procedure of SAS. Variables with $P \leq 0.2$ in the Chi-square analysis (Table 1) were included in the multivariable analyses. The following variables were considered for inclusion in the models: dry period length [**DPL**; ≤ 76 d vs. >76 d; DPL >76 d had the greatest negative effect on reproductive performance in previous studies (Pinedo et al., 2011)], calving season [summer (June–August) and fall (September–November) vs. winter (December–February) and spring (March–May)], calving problems (occurrence of dystocia, twins, stillbirth, abortion, or RP vs. no occurrence), metabolic problems (occurrence of clinical hypocalcaemia or ketosis vs. no occurrence) up to 14 DIM, digestive problems (occurrence of indigestion or DA vs. no occurrence) up to 14 DIM, metritis (yes vs. no) up to 14 DIM, mastitis (yes vs. no) up to 14 DIM, BW loss (**BWL**) within 14 DIM (lost ≤ 28 kg vs. lost >28 kg; 56 kg represents 1 unit of BCS; Otto et al., 1991), average milk yield in the first 14 DIM ($\geq$ average vs. $<$ average; average was 20.5 kg for primiparous and 31.0 kg for multiparous cows), and BCS at 17 $\pm$ 3 DIM (≤ 2.75 vs. >2.75). Two-way interactions between parity and other covariates were included whenever possible. Herd was included as a random variable. Three models were constructed considering availability of daily cow weights and dry period. Therefore, model 1 did not include BWL or DPL (768 cows available), model 2 included BWL in the first 14 DIM and excluded DPL (458 cows available because weights were not recorded for 3 cows in the first 14 d in dairy 1), and model 3 included DPL but excluded BWL (445 multiparous cows available).

Table 1. Descriptive statistics and Chi-square test results for factors associated with cyclicity by 21 ± 3 DIM

Variable	Level	Cows, no.	Cyc21, ¹ %	P-value
Herd	Dairy 1	461	42.3	0.66
	Dairy 2	307	40.7	—
Parity	Multiparous	445	45.4	0.01
	Primiparous	323	36.5	—
Calving season ²	SumFall	598	45.7	<0.001
	WinSpri	170	27.7	—
Calving problem ³	No	544	43.8	0.07
	Yes	224	36.6	—
Metabolic problem ⁴	No	599	43.7	0.03
	Yes	169	34.3	—
Metritis ⁵	No	637	43.5	0.02
	Yes	131	32.8	—
Mastitis ⁶	No	715	41.7	0.98
	Yes	53	41.5	—
Digestive problem ⁷	No	696	42.8	0.04
	Yes	72	30.6	—
Lameness ⁸	No	746	41.7	0.94
	Yes	22	40.9	—
BCS ⁹	>2.75	584	42.8	0.25
	≤2.75	184	38.0	—
Milk yield ¹⁰	≥Average	421	42.0	0.82
	<Average	347	41.2	—
BWL ¹¹	≤28 kg	156	48.7	0.05
	>28 kg	300	39.3	—
DPL ¹²	≤76 d	362	48.3	0.009
	>76 d	83	32.5	—

¹Cyc21 = cyclic by 21 ± 3 DIM.

²Calving season = summer (June–August; n = 436; Cyc21 = 42%), fall (September–November; n = 162; Cyc21 = 55.6%), winter (December–February; n = 76; Cyc21 = 27.6%), spring (March–May; n = 94; Cyc21 = 27.7%); SumFall = summer and fall, WinSpri = winter and spring.

³Calving problem = abortion, dystocia, retained placenta, twins, or stillbirth.

⁴Metabolic problem = ketosis (no = 607; Cyc21 = 43.2%; yes = 161; Cyc21 = 36.0%) or clinical hypocalcemia (no = 752; Cyc21 = 42.2%; yes = 16; Cyc21 = 18.8%) within 14 DIM.

⁵Metritis = fetid red-brownish, watery vaginal discharge within 14 DIM.

⁶Mastitis = abnormal milk or mammary gland within 14 DIM.

⁷Digestive problem = indigestion (diarrhea, decreased rumen motility, or bloat; no = 702; Cyc21 = 42.6%; yes = 66; Cyc21 = 31.8%) or displaced abomasum (no = 758; Cyc21 = 42.0%; yes = 10; Cyc21 = 20.0%) within 14 DIM.

⁸Lameness = favoring 1 of the limbs while walking.

⁹BCS at 17 ± 3 DIM.

¹⁰Milk yield = average was 20.5 and 31.0 kg for primiparous and multiparous, respectively.

¹¹BWL = BW loss within 14 DIM.

¹²DPL = dry period length.

A manual backward elimination was performed, and variables were removed from the model if $P > 0.10$. With a sample size of 768 cows, 80% power, α of 5%, and equal group sizes, differences of 7 percentage units (e.g., 44 vs. 37%) would be significant in a univariable analysis. Nonetheless, when performing a multivariable analysis, adjustments are made for all variables left in the final model; therefore, variables that are significant in a univariable analysis may become nonsignificant and vice versa. Furthermore, unequal group sizes may decrease the power. With the same parameters listed previously but with 1 group 2.5 times larger than the other, the power would decrease to 73% or only differences of 7.7 percentage units would be significant.

With a sample size of 458 cows, 80% power, α of 5%, and equal group sizes, differences of 9 percentage units (e.g., 44 vs. 35%) would be significant in a univariable analysis. Likewise, having 1 group 2.5 times larger than the other the power would decrease to 73% or only differences of 9.8 percentage units would be significant. With a sample size of 445 cows, 80% power, α of 5%, and equal group sizes, differences of 9.1 percentage units (e.g., 44 vs. 34.9%) would be significant in a univariable analysis. Likewise, having 1 group 2.5 times larger than the other the power would decrease to 73% or only differences of 9.9 percentage units would be significant. Correlation between the main variables included in the multivariable analysis was tested with Spearman rank

correlation coefficient using the CORR procedure of SAS. Statistical significance was set at $P \leq 0.05$ and a tendency toward statistical significance was considered when $0.05 < P \leq 0.10$.

RESULTS

Univariable Chi-Square Analysis and Spearman Correlations

Descriptive statistics are shown in Table 1. Overall, cyclicity by 21 ± 3 DIM was 41.2% (320/768). The univariable Chi-squared analyses indicated that cyclicity by 21 ± 3 DIM was significantly associated ($P \leq 0.05$) with calving season, parity, metabolic problems, metritis, digestive problems, BWL, and DPL. Calving problems tended ($P = 0.07$) to be associated with cyclicity by 21 ± 3 DIM. Cyclicity by 21 ± 3 DIM was not associated ($P \geq 0.25$) with herd, mastitis, lameness, or milk yield in the first 14 DIM, or BCS at 17 ± 3 DIM.

The correlation matrix for the variables included in the multivariable models is shown in Table 2. Correlations among variables were tested to help explain the results of the multivariable models. Caving in the winter or spring was positively correlated with being primiparous, having metritis, and having digestive problems within 14 DIM, but was negatively correlated with and with losing >28 kg of BW within 14 DIM. Being primiparous was positively correlated with having calving problems and metritis, and negatively correlated with having metabolic problems and losing >28 kg of BW within 14 DIM. Having calving problems was positively correlated with having metritis and dry period >76 d, and was negatively correlated with having digestive problems within 14 DIM. Having metabolic problems within 14 DIM was positively correlated with having metritis and digestive problems within 14 DIM, with losing >28 kg of BW within 14 DIM, and with having dry period >76 d. Having metritis within 14 DIM was positively correlated with having digestive problems within 14 DIM and with losing >28 kg within 14 DIM. Having digestive problems within 14 DIM was positively correlated with having dry period >76 d.

Multivariable GLIMMIX Model 1

Results from the multivariable mixed logistic regression model including all 768 cows (model 1) are shown in Table 3. Calving season, parity, metabolic problems, and digestive problems had a significant effect ($P \leq 0.05$) on cyclicity by 21 ± 3 DIM. Cows that calved in the summer or fall had greater cyclicity than cows that calved in the winter or spring [45.7 vs. 27.7%; odds ratio (OR) = 2.4; 95% CI = 1.6–3.6; $P < 0.001$], multipa-

Table 2. Correlation matrix showing Spearman's rho correlation coefficient (P -values in parentheses)

Item	Calved in WinSpri ¹	Primiparous ²	Calving problem ³	Metabolic problem ⁴	Metritis ⁵	Digestive problem ⁶	BWL ⁷	DPL ⁸
Calved in WinSpri	1 (—)							
Primiparous		0.19 (<0.001) 1 (—)	0.02 (0.64) 0.15 (<0.001) 1 (—)	0.05 (0.17) -0.22 (<0.001) 0.005 (0.89) 1 (—)	0.12 (0.001) 0.17 (<0.001) 0.13 (<0.001) 0.19 (<0.001) 1 (—)	0.11 (0.002) -0.05 (0.19) -0.09 (0.01) 0.15 (<0.001) 0.09 (0.01) 1 (—)	-0.10 (0.04) -0.24 (<0.001) -0.04 (0.42) 0.29 (<0.001) 0.18 (<0.001) 0.07 (0.15) 1 (—)	0.005 (0.91) — 0.09 (0.05) 0.17 (<0.001) -0.05 (0.30) -0.13 (0.007) 0.08 (0.18) 1 (—)
Calving problem								
Metabolic problem								
Metritis								
Digestive problem								
BWL								
DPL								

¹Cows that calved in the winter (December–February) and Spring (March–May) were coded as 1 and cows that calved in the summer (June–August) and fall (September–November) were coded as 0; WinSpri = winter and spring calving.

²Primiparous cows were coded as 1 and multiparous were coded as 0.

³Cows that had calving problems (abortion, dystocia, retained placenta, twins, or stillbirth) were coded as 1 and cows that did not have calving problems were coded as 0.

⁴Cows that had metabolic problems (ketosis or clinical hypocalcemia within 14 DIM) were coded as 1 and cows that did not have metabolic problems were coded as 0.

⁵Cows that had metritis (fetid red-brownish, watery vaginal discharge within 14 DIM) were coded as 1 and cows that did not have metritis were coded as 0.

⁶Cows that had digestive problems (diarrhea, decreased rumen motility, bloat, or displaced abomasum within 14 DIM) were coded as 1 and cows that did not have digestive problems were coded as 0.

⁷BWL = BW loss within 14 DIM. Cows that lost >28 kg were coded as 1 and cows that lost ≤ 28 kg were coded as 0.

⁸DPL = dry period length. Cows with DPL >76 were coded as 1 and cows with DPL ≤ 76 were coded as 0.

Table 3. Final multivariable logistic regression model (model 1) for factors associated with cyclicity by 21 ± 3 DIM including all cows ($n = 768$)

Variable	Level	Cows, no.	Cyc21, ¹ %	Odds ratio	95% CI	<i>P</i> -value
Calving season ²	SumFall	598	45.7	2.4	1.6–3.6	<0.001
	WinSpri	170	27.7	Ref. ³	Ref.	—
Parity	Multiparous	445	45.4	1.5	1.1–2.1	0.01
	Primiparous	323	36.5	Ref.	Ref.	—
Metabolic problem ⁴	No	599	43.7	1.7	1.2–2.6	0.005
	Yes	169	34.3	Ref.	Ref.	—
Digestive problem ⁵	No	696	42.8	1.8	1.0–3.1	0.05
	Yes	72	30.6	Ref.	Ref.	—

¹Cyc21 = cyclic by 21 ± 3 DIM.²Calving season = summer (June–August), fall (September–November), winter (December–February), spring (March–May); SumFall = summer and fall calving, WinSpri = winter and spring calving.³Ref. = Referent.⁴Metabolic problems = ketosis or clinical hypocalcemia within 14 DIM.⁵Digestive problems = indigestion (diarrhea, decreased rumen motility, or bloat) or displaced abomasum within 14 DIM.

rous cows had greater cyclicity than primiparous cows (45.4 vs. 36.5; OR = 1.5; 95% CI = 1.1–2.1; $P = 0.01$), cows that did not have metabolic problems had greater cyclicity than cows that had metabolic problems (43.7 vs. 34.3%; OR = 1.7; 95% CI = 1.2–2.6; $P = 0.05$), and cows that did not have digestive problems had greater cyclicity than cows that had digestive problems (42.8 vs. 30.6%; OR = 1.8; 95% CI = 1.0–3.1; $P = 0.05$). Two-way interactions between parity and other covariates were not significant ($P > 0.10$).

Multivariable GLIMMIX Model 2

Results from the multivariable mixed logistic regression model including only cows from dairy 1 ($n = 458$; model 2) are shown in Table 4. Calving season, metritis, and digestive problems had a significant effect ($P \leq 0.05$) on cyclicity by 21 ± 3 DIM. A tendency ($P = 0.06$) was noted for an effect of BWL within 14 DIM on cyclicity by 21 ± 3 DIM. Cows that calved in the summer or fall had greater cyclicity than cows that calved in the winter or spring (50.0 vs. 28.5%; OR = 2.7; 95% CI = 1.7–4.1; $P < 0.001$), cows that did not have metritis had greater cyclicity than cows that had metritis (46.2 vs. 31.6%; OR = 1.7; 95% CI = 1.1–2.8; $P = 0.02$), and cows that did not have digestive problems had greater cyclicity than cows that had digestive problems (45.0 vs. 29.6%; OR = 1.9; 95% CI = 1.1–3.3; $P = 0.03$). Cows that lost ≤ 28 kg within 14 DIM tended to have greater cyclicity than cows that lost > 28 kg within 14 DIM (49.0 vs. 39.2%; OR = 1.5; 95% CI = 1.0–2.3; $P = 0.06$). Two-way interactions between parity and other covariates were not significant ($P > 0.10$).

Multivariable GLIMMIX Model 3

Results from the multivariable mixed logistic regression model including only cows with a dry period ($n = 445$; model 3) are shown in Table 5. Metabolic problems and DPL had a significant effect ($P = 0.03$) on cyclicity by 21 ± 3 DIM. Cows that did not have metabolic problems had greater cyclicity than cows that had metabolic problems (49.4 vs. 36.1%; OR = 1.6; 95% CI = 1.0–2.5; $P = 0.03$) and cows that had a DPL ≤ 76 d had greater cyclicity than cows that had a DPL > 76 d (48.3 vs. 32.5%; OR = 1.8; 95% CI = 1.1–3.0; $P = 0.03$). Two-way interactions between parity and other covariates were not tested because only multiparous cows were included in the analysis.

DISCUSSION

In our study, 41.2% of all cows were cyclic by 21 ± 3 DIM. This is within reported ranges (Darwash et al., 1997; McCoy, 2006; Kim et al., 2012), although it is higher than 2 recent studies in North America which reported cyclicity of 26 (Galvão et al., 2010a) and 28% (Dubuc et al., 2012) at 21 ± 3 DIM. The reason for this difference might be partly explained by the fact that, in the studies by Galvão et al. (2010a) and Dubuc et al. (2012), progesterone ≥ 1 ng/mL was used to determine cyclicity at 21 ± 3 DIM whereas herein we used US to detect the presence of a CL in the ovaries. Although it takes 3 to 4 d after ovulation for progesterone concentrations to reach 1 ng/mL (Sartori et al., 2004; Wolfenson et al., 2004; Kayacik et al., 2005), a CL can be observed the day of or a day after ovulation (Kastelic et al., 1990; Kayacik et al., 2005); therefore, using progesterone concentrations might slightly underestimate

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Table 4. Final multivariable logistic regression model (model 2) for factors associated with cyclicity by 21 ± 3 DIM including cows that had BW measured (only dairy 1; $n = 458$)

Variable	Level	Cows, no.	Cyc21, ¹ %	Odds ratio	95% CI	P-value
Calving season ²	SumFall	300	50.0	2.7	1.7–4.1	<0.001
	WinSpri	158	28.5	Ref. ³	Ref.	—
Metritis ⁴	No	344	46.2	1.7	1.1–2.8	0.02
	Yes	114	31.6	Ref.	Ref.	—
Digestive problem ⁵	No	387	45.0	1.9	1.1–3.3	0.03
	Yes	71	29.6	Ref.	Ref.	—
BWL ⁶	≤28 kg	157	49.0	1.5	1.0–2.3	0.06
	>28 kg	301	39.2	Ref.	Ref.	—

¹Cyc21 = cyclic by 21 ± 3 DIM.²Calving season = summer (June–August), fall (September–November), winter (December–February), spring (March–May); SumFall = summer and fall calving, WinSpri = winter and spring calving.³Ref. = referent.⁴Metritis = fetid red-brownish, watery vaginal discharge within 14 DIM.⁵Digestive problems = indigestion (diarrhea, decreased rumen motility, or bloat) or displaced abomasum within 14 DIM.⁶BWL = BW loss within 14 DIM.

cyclicity by 21 ± 3 DIM. Another important factor was the fact that most of the cows (78%) enrolled in our study had calved in the summer or fall, and those cows had much greater cyclicity by 21 ± 3 DIM than cows that calved in the winter and spring (45.7 vs. 27.7%). Other factors that were negatively associated with cyclicity by 21 ± 3 DIM were being primiparous, having metritis, metabolic or digestive problems, losing >28 kg of BW and having dry period >76 d; each will be discussed herein.

Calving season was the most significant variable in models 1 and 2. Several studies evaluated the effect of season on cyclicity; some have observed greater cyclicity for cows calving in fall and summer (Walsh et al., 2007; Santos et al., 2009; Kim et al., 2012), some for cows calving in spring and summer (El-Din. Zain et al., 1995; Dubuc et al., 2012), and some for cows calving in spring and fall (Opsomer et al., 2000), but none have observed greater cyclicity for cows calving in winter. Photoperiod may affect follicle development because melatonin is decreased with increased day length, and

melatonin has been shown to decrease LH pulsatility in ovariectomized cows (Rhodes et al., 1979) and to suppress IGF-1 (Dahl et al., 2000), which stimulates estradiol production and follicle growth (Beam and Butler, 1998; Butler, 2003). In ovariectomized heifers, it has been observed that release of LH induced by estradiol was greater for heifers exposed to 18 h of light and 6 h of dark than for heifers exposed to short days (Hansen et al., 1982). Furthermore, primiparous cows that calved in the fall and winter but were exposed to 18 h of light and 6 h of dark had reduced intervals to first ovulation and to first estrus compared with cows exposed to natural day light (Hansen and Hauser, 1984). Therefore, it is reasonable to expect that cows exposed to more light during the spring, summer, or fall would have greater cyclicity than cows that calve in the winter.

Metabolic problems were significantly associated with cyclicity by 21 ± 3 DIM in models 1 and 3. Furthermore, BWL, which was highly positively correlated with metabolic problems, was associated with cyclicity

Table 5. Final multivariable logistic regression model (model 3) for factors associated with cyclicity by 21 ± 3 DIM including only multiparous cows ($n = 455$)

Variable	Level	Cows, no.	Cyc21, ¹ %	Odds ratio	95% CI	P-value
Metabolic problem ²	No	312	49.4	1.6	1.0–2.5	0.03
	Yes	133	36.1	Ref. ³	Ref.	—
DPL ⁴	≤76 d	362	48.3	1.8	1.1–3.0	0.03
	>76 d	83	32.5	Ref.	Ref.	—

¹Cyc21 = cyclic by 21 ± 3 DIM.²Metabolic problems = ketosis or clinical hypocalcemia within 14 DIM.³Ref. = referent.⁴DPL = Dry period length.

by 21 ± 3 DIM in model 2. Early studies by Beam and Butler (1997, 1998) observed that cows that ovulated their dominant follicle of the first postpartum follicular wave had a lesser degree of negative energy balance and especially a shorter interval to the nadir of negative energy balance (Beam and Butler, 1997, 1998). Deep negative energy balance affects LH pulsatility, and blood glucose, insulin, and IGF-1 concentrations, which ultimately affect follicle development, estradiol production, and ovulation (Canfield and Butler, 1990; Beam and Butler, 1999; Butler, 2003). Ketosis and BWL are indicators of negative energy balance, as cows need to mobilize body fat in the form of NEFA to meet the energy demands for maintenance, growth, and milk production, which lead to BWL and production of BHBA in the liver (Vazquez-Añón et al., 1994; Butler, 2003). Herein we grouped hypocalcemia and ketosis as metabolic problems because both are associated with a decrease in DMI and an increase in NEFA and BHBA (Martinez et al., 2014). In fact, 50% (8/16) of the cows that had clinical hypocalcemia also had ketosis. Our findings agree with previous findings in which high NEFA pre- and postpartum was negatively associated with cyclicity at 21 ± 3 DIM (Dubuc et al., 2012) and in which subclinical or clinical ketosis was negatively associated with cyclicity by 50 to 60 DIM (Opsomer et al., 2000; Garbarino et al., 2004; Walsh et al., 2007). Hypocalcemic cows have also been found to have a longer interval from calving to their first ovulation postpartum (Risco et al., 1994). When evaluated as a continuous variable, BWL within 30 DIM was positively correlated with time to resumption of ovulation ($r = 0.736$, $P < 0.001$), meaning that cows that lose more BW take longer to resume cyclicity (El-Din. Zain et al., 1995). Others have looked at BCS change as a proxy for BWL and have found a significant negative association between losing BCS in the first 30 to 65 DIM and cyclicity by 45 to 65 DIM (Opsomer et al., 2000; Shrestha et al., 2005; Santos et al., 2009).

Digestive problems were significantly associated with cyclicity by 21 ± 3 DIM in models 1 and 2. The most common cause of digestive problems was probably SARA, because of all the cows classified as having digestive problems ($n = 72$), 66 had only indigestion, 10 had only DA, and 4 had both. The signs of indigestion used herein (diarrhea, decreased rumen motility, or bloat) are characteristic of SARA (Plaizier et al., 2008). Digestive problems may affect early ovulation through negative effects on energy balance or through inflammation. Herein, digestive problems were highly positively correlated with metabolic problems; therefore, both may affect early ovulation by negatively affecting energy balance. When cows are induced to develop SARA by feeding high-concentrate diets, a

severe drop in DMI is observed (Plaizier et al., 2008; Khafipour et al., 2009). Displaced abomasum also has a strong association with negative energy balance (McArt et al., 2013). Several studies in which SARA was induced in dairy cows have consistently shown an increase in the concentrations of LPS originating from gram-negative bacteria in rumen fluid (Andersen et al., 1994; Khafipour et al., 2009; Li et al., 2012) and also in blood (Khafipour et al., 2009). Lipopolysaccharide has been shown to impair the release of both GnRH and LH (Peter et al., 1989; Rivest et al., 1993; Battaglia et al., 1999) and to decrease aromatase activity (Herath et al., 2009), which ultimately may affect follicular growth, estradiol production (Williams et al., 2007, 2008), and decrease ovulation rate (Peter et al., 1989). To our knowledge, only the association between DA and cyclicity has been evaluated. Walsh et al. (2007) observed that cows that had DA had decreased cyclicity by 60 DIM. Others have found that cows with a DA had more days from calving to insemination and from calving to conception (Pedersen, 2006).

Metritis was negatively associated with early cyclicity. To our knowledge, this is the first time metritis has been associated with delayed cyclicity. Others have found a negative effect of abnormal vaginal discharge (probably endometritis) and abnormal calvings (dystocia, abortions, or twins) on cyclicity by 50 DIM (Opsomer et al., 2000) or on time to resumption of ovarian cyclicity (El-Din. Zain et al., 1995). Likewise, Walsh et al. (2007) found a negative effect of twins, dystocia, and RP on cyclicity by 60 DIM. Metritis was offered in the multivariable model in the study by Opsomer et al. (2000), but not in the study by Walsh et al. (2007). Others have observed that haptoglobin, an indicator of inflammation increased in cows with metritis (Huzzey et al., 2009; Galvão et al., 2010b), was associated with decreased cyclicity by 21 ± 3 DIM (Dubuc et al., 2012). Given the association of risk factors for metritis, such as dystocia, abortions, twins, RP, and delayed cyclicity, it makes sense that metritis would be associated with delayed cyclicity. The negative effect of metritis on cyclicity by 21 ± 3 DIM may be partly explained by the association between metritis and negative energy balance (Hammon et al., 2006; Huzzey et al., 2007; Galvão et al., 2010b) and by the release of LPS into the uterine lumen, which end up in the blood circulation (Mateus et al., 2003; Herath et al., 2009). As mentioned above, LPS can affect GnRH and LH release, follicle development, estradiol production, and ovulation.

Primiparity has been associated with delayed cyclicity in 2 previous studies (Santos et al., 2009; Kim et al., 2012); however, 1 study observed that primiparous cows had increased cyclicity compared with cows with ≥ 3 lactations (Dubuc et al., 2012). Herein, primiparity

was positively correlated with calving problems and metritis, but negatively correlated metabolic problems and BWL, therefore, making it difficult to identify the causes on decreased cyclicity in primiparous cows because a higher incidence of metritis would have a detrimental effect on early cyclicity but a lower incidence of metabolic problems would have a positive effect on early cyclicity. Primiparous cows are in a completely different metabolic state compared with multiparous because they are lactating while still growing; therefore, an even a lower degree of negative energy balance may have a greater negative effect on the hormonal milieu controlling cyclicity.

The DPL has been previously identified as a risk factor for delayed cyclicity. Opsomer et al. (2000) observed that cows with a dry period >77 d had delayed cyclicity by 50 DIM. Others have observed increased calving to first service interval, calving to conception interval, and number of services per conception in cows with DPL >76 d (Pinedo et al., 2011). Herein, DPL >76 d was positively correlated with calving problems and metabolic problems but negatively correlated with digestive problems. The reason for a long dry period is usually an early dry-off because of low milk production at the end of lactation in cows that require a longer time to become pregnant. Using data from 1,006 cows from a recent study (Gobikrushanth et al., 2014), it was observed that mean time to conception (days open) was 42 d longer in cows with DPL >76 d compared with cows with DPL ≤76 d (176 vs. 134; $P < 0.001$). Therefore, cows with DPL >76 d would have gained more BW by the end of the lactation because of the longer lactation and by the time of calving because of longer DPL, which would predispose them to calving and metabolic problems which would then predispose them to delayed cyclicity because of greater negative energy balance (Morrow, 1976). Indeed, data from the current study showed that cows with DPL >76 d had greater BCS at 17 ± 3 DIM than cows with DPL ≤76 d (3.25 vs. 3.08; $P < 0.001$). It is important to note that although having 3 different models allowed us to focus on important risk factors, such as BWL (model 2) and DPL (model 3), it also caused a loss of power as presented in the statistical analysis section, which could have caused some variables that were significant in model 1 to become nonsignificant in models 2 or 3.

CONCLUSIONS

In summary, cyclicity by 21 ± 3 DIM was negatively associated with calving in winter or spring, primiparity, metritis, metabolic or digestive problems, >28 kg of BWL, and a DPL >76 d. Strategies preventing extended DPL and BWL, together with reductions in

the incidence of metritis and metabolic and digestive problems, should improve early cyclicity postpartum.

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